

Appendix S1 – Supplementary Information

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Beaver-engineering amplifies dissolved organic carbon concentration

This document contains the material supporting the main findings of the manuscript (tested hypotheses of the structural causal model, priors, residuals, posterior predictive checks and posterior parameter distributions). The full description of data collection, variable derivation and model specification is given in the Methods of the main manuscript.

SCM: tested hypothesis

Table S1

Table S1. Tested hypothesis for the structural causal model (SCM) represented in Figure 1b, main document.

Causal path	Predictor		Response	Hypothesis
1	DOC up-stream		ΔDOC	If DOC upstream increases, then ΔDOC decreases because beaver ponds can act as sink, retaining or transforming DOC through processes such as microbial degradation, sedimentation, and photodegradation, thereby reducing downstream DOC concentrations relative to upstream inputs (Larson et al. 2007; Wohl 2013; Johnston 2014).
2	Macrophytes		ΔDOC	If macrophyte abundance increases, we expect ΔDOC to increase because the decomposition of macrophyte biomass releases labile dissolved organic carbon, enhancing downstream concentrations (Reitsema et al. 2018).
3	Phytoplankton		ΔDOC	If phytoplankton abundance increases, then ΔDOC increases is likely to increase, because phytoplankton contribute to DOC content through exudation during growth and release upon cell lysis, thereby enhancing downstream DOC levels relative to upstream inputs (Søndergaard et al. 1985; Sell and Overbeck 1992).
4	Soil cover	litter	ΔDOC	If soil litter cover increases, then ΔDOC increases as decomposition of litter releases dissolved organic compounds into the water (Aitkenhead et al. 1999; Cincotta et al. 2019).
5	Water residence time	resi-	ΔDOC	If water residence time increases, then ΔDOC increases, because longer water residence times allow for more extensive microbial decomposition and processing of dissolved organic carbon, enhancing downstream DOC concentrations (Jepsen et al. 2019; Abbasi et al. 2024).
6	Solar radiation	radia-	Phytoplankton	If solar radiation increases then phytoplankton abundance is expected to increase because greater light availability enhances photosynthetic rates (Reynolds 2006).
7	Solar radiation	radia-	Macrophytes	If solar radiation increases, then macrophyte abundance increases because higher light availability enhances photosynthetic activity and supports plant growth (Feuchtmayr et al. 2009).
8	Solar radiation	radia-	Litter cover	If solar radiation increases, as in summer, then soil litter cover decreases because lower rates of tree leaf fall coincide with enhanced decomposition driven by higher temperatures and solar radiation (Gessner et al. 1999).
9	Water residence time	resi-	Macrophytes	If water residence time increases, macrophyte abundance may increase

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Structural Causal Model: the priors

To ensure the model was not over-constrained by prior knowledge, we used weakly informative priors for all parameters. This approach allows the data to inform the model without assuming strong beliefs about the parameter values beforehand (Figure S1). We note, however, that for macrophytes, phytoplankton, and litter cover—variables measured in 16 of the 176 streams—we used more informative priors that incorporate established ecological knowledge about their influence on DOC concentrations (Figure S1 panel b).

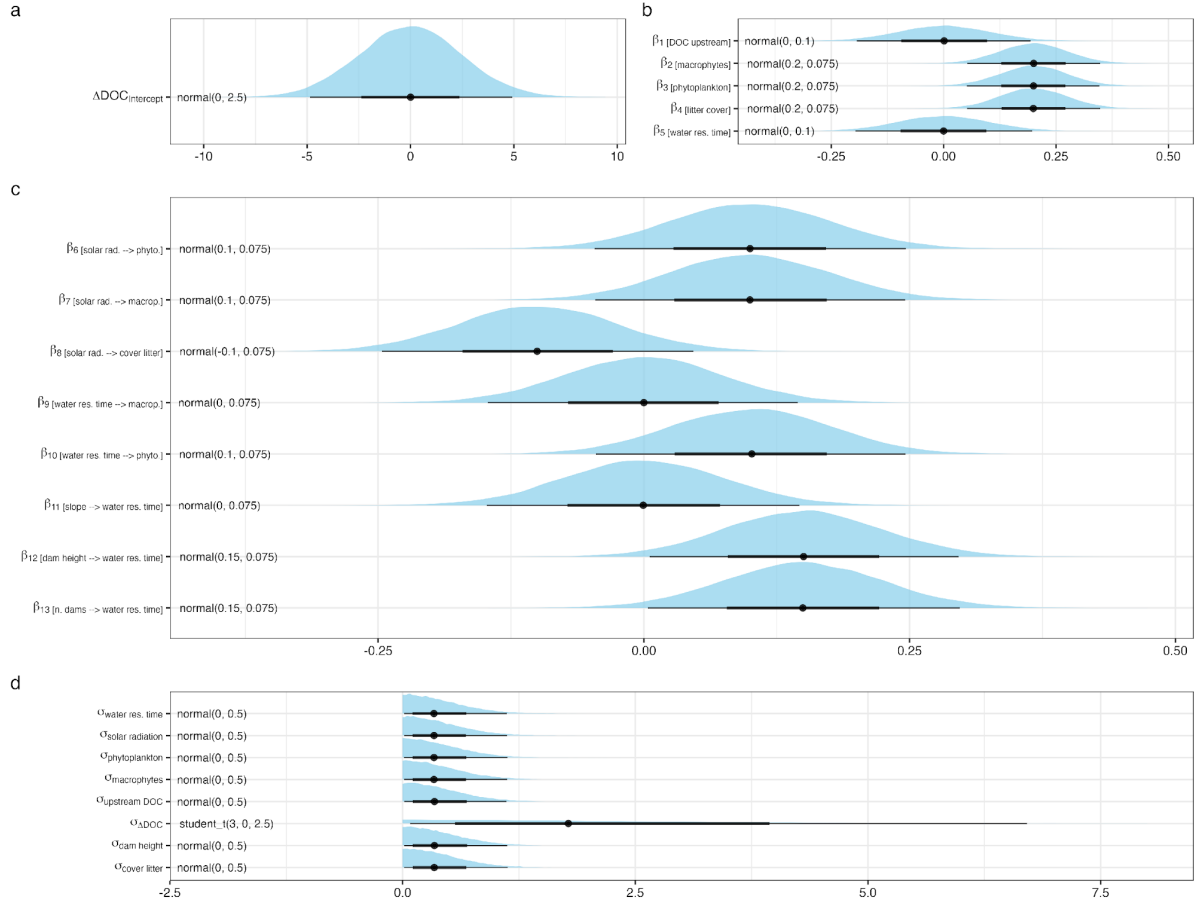


Figure S1: Half-eye plots display the prior distributions used in the model. The shaded areas (slabs) represent the probability density of each prior distribution, illustrating their shape and spread. The central dot indicates the median of the distribution, while the thick and thin horizontal lines represent the 66% and 95% credible intervals, respectively. Text labels to the left of each distribution indicate the corresponding prior specification.

Structural Causal Model: the residuals

We can evaluate the model fit by looking at the difference between observed data and their predictions. That are the residuals (r_i).

$$r_i = y_i - y_i^{\text{pred}} \quad (\text{S1})$$

If the model is a reasonable representation of the observational process that originates the data then the residuals should look roughly randomly scattered in comparison to a horizontal line.

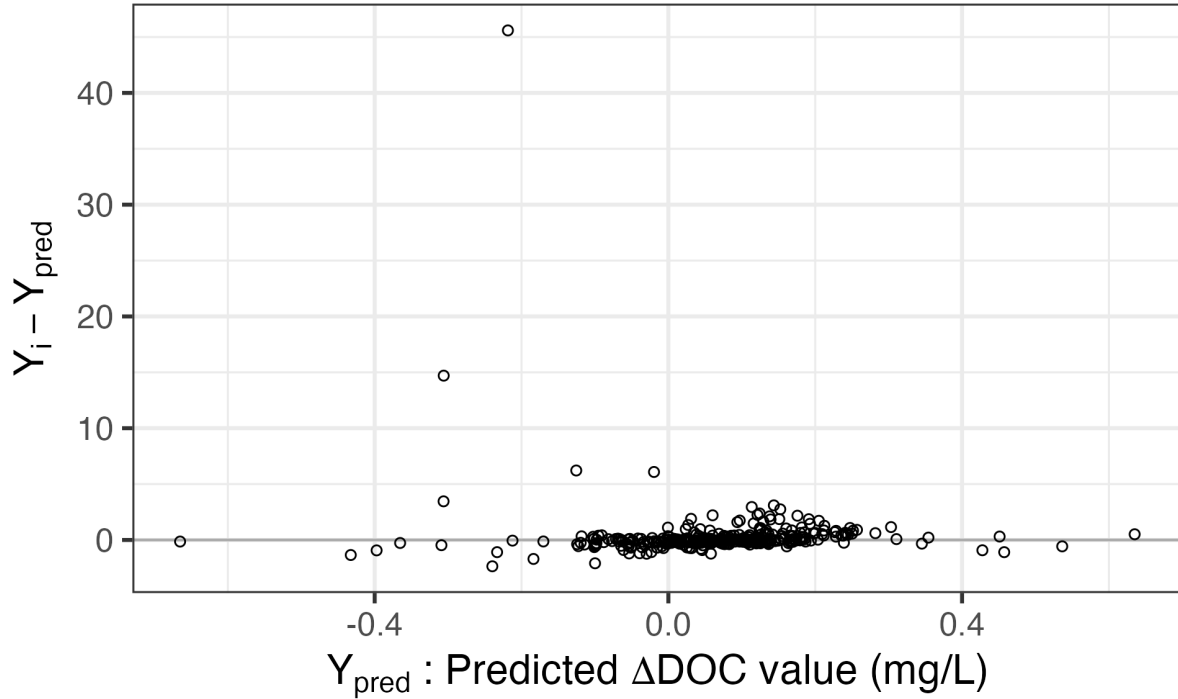


Figure S2: Residual plot showing the relationship between model-predicted ΔDOC values (mg L^{-1}) and corresponding residuals. Points represent the difference between observed and predicted values; a random scatter around zero (grey line) indicates good model fit and no systematic bias in predictions.

Structural Causal Model: posterior predictive checks

Posterior predictive checks assess how well the fitted SCM model reproduces observed data by comparing the observed values to data simulated from the model's posterior predictive distribution.

Structural Causal Model: posterior parameters distribution

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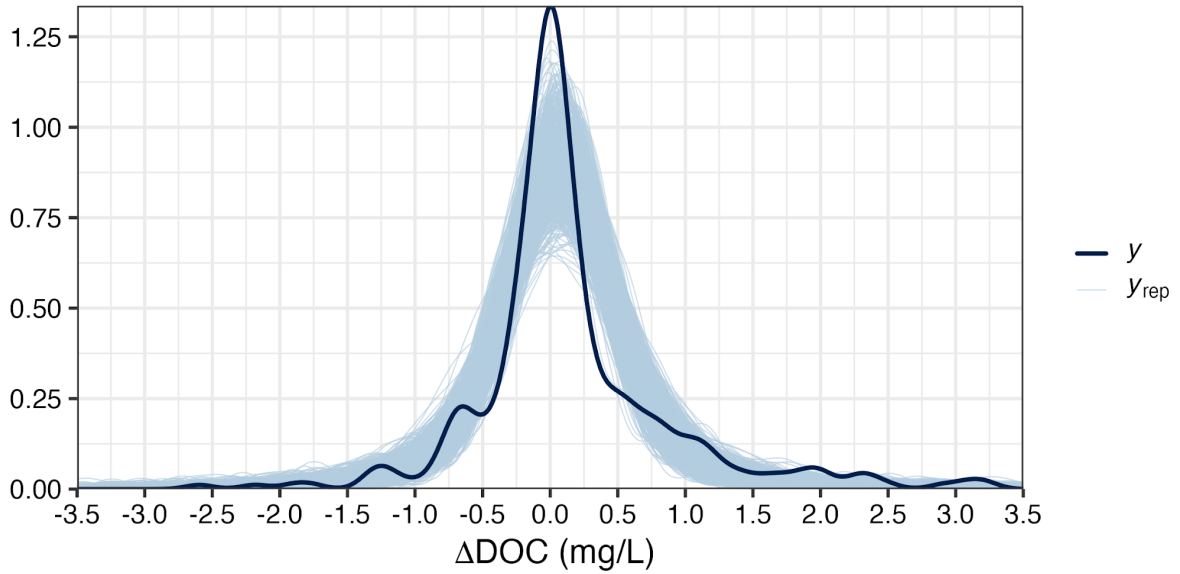


Figure S3: Posterior predictive checks of the mean and standard deviation statistics for the response variable delta DOC, with 1000 posterior draws used to generate the predictive distribution. This plot provides a visual assessment of the model's fit by comparing the observed data (dark blue) against the model's predictions (light blue) from the posterior distribution.

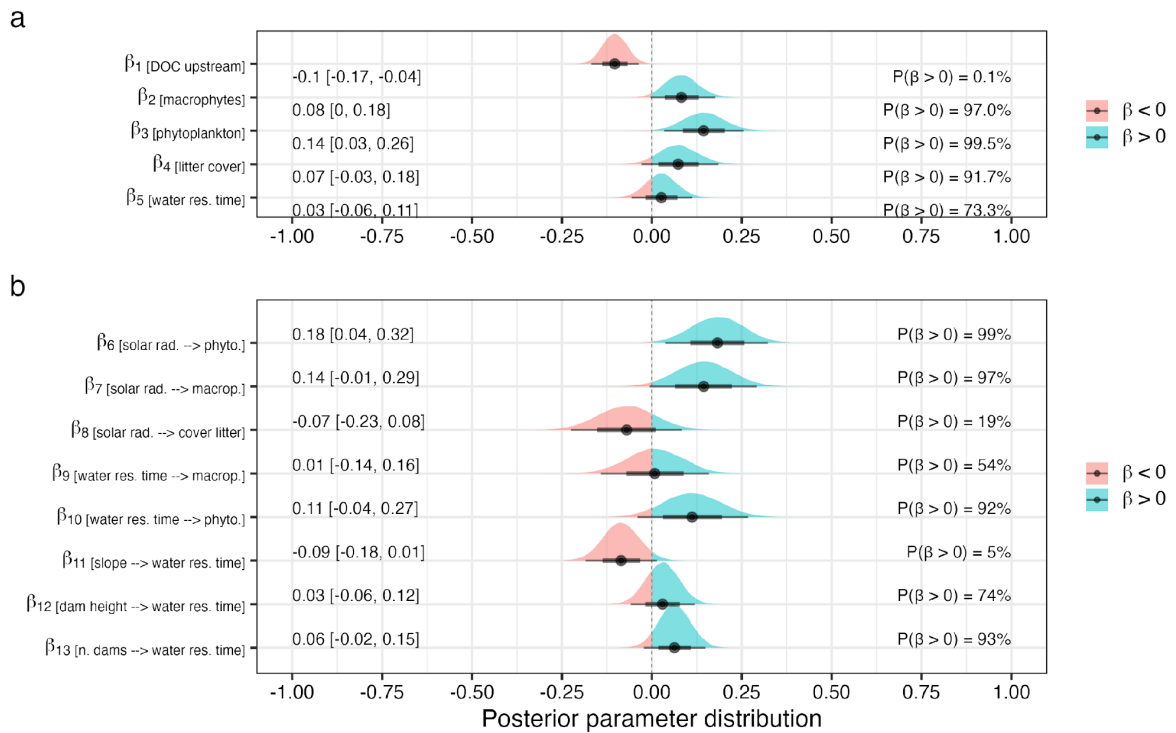


Figure S4: Posterior parameters distributions of the fitted SCM. Median shown with point; 66% and 95% credible intervals shown with black bars. Parameters that have more than 90% probability mass on either side of zero are considered to have a positive or negative influence on the response variable. a) The direct effects on ΔDOC b) The indirect effects on ΔDOC .

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